International Social Science Journal

WILEY

INTERNATIONAL SOCIAL SCIENCE JOURNAL

ARTICLE OPEN ACCESS

Female Literacy and Crime Against Women in India: Complex Trajectories and Correlations

Suzana Košir[1] | Radhika Lakshminarayanan[2]

[1]Faculty of Arts, University of Maribor, Maribor, Slovenia | [2]Independent Researcher

Correspondence: Suzana Košir (suzana.kosir1@um.si)

Received: 12 September 2024 | **Revised:** 29 October 2024 | **Accepted:** 20 November 2024

Funding: The authors received no specific funding for this work.

Keywords: crime against women | education | female literacy | legislation | women's empowerment

ABSTRACT

Gender equality remains a recurrent goal of governments across the world, yet discrimination and abuse continue to impede the progress of women. Within India's patriarchal society, women continue to be viewed through a stereotypical lens. Despite greater focus on female literacy and enactment of legislations that seek to protect women's rights, the safety of women from crime remains tenuous. This research explores trajectories of female literacy and crimes against women (CAW) in India to find possible correlations. Data from the Ministry of Statistics, the National Crime Record Bureau, National Census records and official sample surveys were obtained from government databases. Further, bivariate regression analysis was used to validate the hypothesis. The results indicate a clear correlation between female literacy and CAW, although other factors may be considered to explain certain anomalies.

1 | Introduction

Today's world is increasingly knowledge-driven, and literacy is necessary in empowering citizens to critique injustice and create new pathways for sustainable development. Particularly in the case of women's empowerment, literacy is a critical component that not only transforms women's lives but also contributes to creating safer communities by diminishing the root causes of inequality, suppression and oppression. It enables women to make independent and informed decisions, seek financial independence and participate as equal citizens in socio-political and economic realms. When women have equal opportunities, they are better positioned to advocate for themselves and their families by fostering values of equality, fairness and justice, thereby reducing their vulnerability and susceptibility to crime, violence and exploitation. Therefore, women's empowerment is not just a societal imperative but can be a crucial strategy for women's safety and well-being. It is a key factor enabling women to challenge traditional gender roles, leading to more equitable opportunities, less discrimination and better respect within societies.

Violence negatively impacts women and restrains them from holistic participation in society, with ramifications for health care, childcare, family function and loss in economic productivity. Crimes against women (hereafter CAW) threaten their safety and hinder their pursuit of education and careers. Through the Sustainable Development Goal (hereafter SDG) 5, the UN strives to achieve gender equality and women's empowerment. SDG 5.2 aims to 'eliminate all forms of violence against women and girls in the public and private spheres, including trafficking, sexual abuse and other types of exploitation' (World Health Organization (WHO) 2023). However, CAW is an underlying factor that threatens and jeopardises women from enjoying holistic empowerment.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2024 The Author(s). *International Social Science Journal* published by John Wiley & Sons Ltd.

In India, since independence in 1947, the government has attempted to foster an environment of learning through literacy and education. According to the National Family Health report (NFHS 2023), the literacy rate (LR) in 2023 was 77.7%, and female literacy was 70.30% in India. Compared to India's population of 1.4 billion, 1 in 4 people are still below literacy levels. India has repeatedly committed to upholding women's empowerment through equal opportunity; however, despite significant legislation and regulation, women still grapple with deeply rooted patriarchal attitudes and socio-cultural norms that stimulate the perpetuation of CAW. This includes verbal and non-verbal harassment, domestic violence, sexual abuse, dowry, human trafficking, female infanticide and gender-based selective abortions, among others.

Although it is easy to dismiss CAW as an age-old problem related to patriarchal attitudes and functional social norms existing across the globe, the immense increase in such crimes and the more violent and brutal nature of such crimes is disturbing and warrants deeper analysis. Is it more in urban or rural societies? Is its occurrence more common in democratic or authoritarian societies? Does it exist in traditional and conservative societies or more liberal and modern ones? Is women's empowerment a factor that reduces crime? This article is situated on the premise that a deeper analysis of these questions could provide insights into reasons for the increase in CAW and how it can be redressed. The authors were driven to consider if an increase in female literacy impacted the prevalence of CAW. We chose India, a nation with heterogeneous economic, geographic and social settings, to analyse trajectories of female literacy and CAW at the national and state levels to validate possible correlations. Starting with a detailed review of crucial literature on the subject, the article moves on to literacy growth as a marker for women's empowerment in India. It then highlights relevant data on CAW in India from official sources before presenting the methodology to test trajectories of total LR, female LR (FLR) and CAW, followed by a critical discussion of results to validate the research questions.

2 | Literature Review

The definition, concept and interpretations of empowerment are key factors for development policy and practice against inequality and oppression (Rowlands 1995) particularly in the case of women (Rowlands 1998). Kabeer (1999, 2019) locates women at the intersection of class, caste, race, ethnicity, legal and socio-economic status inequalities and defines women's empowerment as the capacity to make choices and live a life outside the one imposed by social laws, customs and norms—women at the intersection. Further research studies the interrelationships between gender equity and empowerment and the measurement of women's empowerment, dimensions and consequences of women's empowerment (Priya, Venkatesh, and Shukla 2021).

There has been considerable academic research on the positive role of literacy in women's empowerment (Fleming 2000; Gallaway and Bernasek 2004; Hill and King 1995; Olson-Strom and Rao 2020). Further research also points to CAW as a hindrance to the development of society (Gill and Mason-Bish 2013; Himabindu, Arora, and Prashanth 2014; Htun and Jensenius 2020; Vicente et al. 2020). A UK-based report studied the impact of low literacy on employment, economic outcomes, health inequalities and crime and concluded that improvements in literacy can create positive, long-lasting change in all aspects of women's development (Morrisroe 2014). Further, Xie, Heimer, and Lauritsen (2012) found that an increase in women's labour force participation, income and education in the US is associated with a decrease in intimate partner violence. Studying demographic risk factors impacting violence against women in Australia, O'Donnell, Smith, and Madison (2002) observed that risks varied with levels of post-school education, income, ethnic background, number and age of children, marital status and age. Thus, scholars have linked women's empowerment to improved economic status, employment, health and education.

Testing the feminist theory of violence against women, Yodanis (2004) posited that the educational and occupational status of women relates to the prevalence of sexual violence against them. Tenkorang et al. (2013) used data from the Ghana Demographic and Health Survey to examine the causes of domestic violence among married women. They concluded that education had a more substantial relation to gender violence than wealth or employment. Studying social inequalities in intimate partner violence among women in Kenya, Lawoko et al. (2007) noted that high literacy among women reduced CAW; however, having a higher education and employment status than the male partner, age differences between partners and lack of autonomy in decision-making increased their vulnerability considerably. This was supported by Deyessa et al. (2010), who explored violence against women in low-income settings in Ethiopia and determined that although semi-urban lifestyle and literacy promoted changes in attitudes against intimate partner violence, in rural settings, a literate woman married to an illiterate man was most vulnerable to spousal violence. Marium (2014) investigated the association between women's education and domestic violence in rural Bangladesh and observed that the educational level of women presented an environment for lower levels of domestic violence and more credible women's autonomy. Another study determined that CAW negatively impacts girls' access to school experience in Malawi (Bisika, Ntata, and Konyani 2009).

Researchers also observed that domestic violence was a result of male controlling behaviours rather than higher female literacy. Estimating the causal effects of education on the prevalence of domestic violence in Turkey, Erten and Keskin (2018) observed that an increase in education among rural women led to an increase in self-reported psychological violence and financial control behaviour without changes in physical violence. A quasi-experimental study conducted in Tehran to determine the effects of teaching about the prevention of domestic violence revealed that it had no effect at all (Noughani and Mohtashami 2011), whereas Jahromi et al. (2016) suggested that raising the level of education of men and women could reduce the risk of domestic violence.

Ross (2006) was among the first to compare the status of women in the Indian state of Kerala and Finland, both highly literate societies. He argued that high female status contributes to improved LRs and life opportunities, whereas high FLRs increase women's social standing. In Indian contexts, scholars have tried to determine other factors that might contribute to violence against women. Drèze and Khera (2000) correlated the female–male ratio

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

in India and suggested that gender relations have a crucial bearing on criminal violence. Another study deduced that an increase in male literacy showed a decline in dowry cases but had no impact on rape, domestic violence and overall crime rates, which indicates that violence against women is deep-rooted in society and relates more to gender-based inequalities (A. Sharma et al. 2020). In a recent study, Lolayekar, Desouza, and Mukhopadhyay (2022) explored cases of rape, kidnapping, cruelty, dowry deaths and molestation across India from 1991 to 2011. They inferred that FLRs, female paid workforce participation and female–male ratio have non-linear effects on CAW.

Attempting to link determinants of women's empowerment and CAW in India, Kumar and Kuncharam (2020) argued that CAW will likely fall if female literacy and workforce participation increase. Further, using data across India, Hazra (2020) surmised that population density, income inequality, poverty and LRs exhibit a significant relationship with crime rates. Simister (2018) points out that many cases of domestic violence arise when the wife challenges male spousal dominance. Through multilevel logistic regression, Das and Basu Roy (2020) demonstrated that women from the poorest economic backgrounds, living in rural areas with low levels of education or no education, were most vulnerable to spousal violence. Exploring the gender gap in literacy (M. Sharma and Kumar 2020) concluded that societal stances, religion and cultural stereotypes directly and indirectly impact literacy.

Analysing patterns of CAW to identify the hotspot areas with high-intensity crime in Rajasthan, Saravag and Kumar (2024a) concluded that likely clusters of CAW are distributed distinctly across years and share boundaries with adjacent states, pointing to the need for collaboration between neighbouring states to optimise their scarce resources to curb such incidents effectively. Further, Saravag and Kumar (2024b) performed a statistical analysis using a composite crime index rooted in diverse categories focusing on health, socioeconomic status, education and judiciary. They concluded that social factors significantly contributed to the occurrence of violence against women.

The above review notes a dearth of research explicitly checking for a correlation between female literacy and CAW at India's national and state levels. Conceptualising from the objectives of this article and after identifying these gaps in the literature, the following research questions were posited:

1. What is the correlation between India's total LR and CAW?
2. What is the correlation between female literacy and CAW in India?
3. What are the trajectories of FLRs and CAW in India's states and union territories?
4. What is the significance of the correlation between female literacy and CAW? Are there any anomalies?

Further, the authors seek to identify anomalies in specific states and suggest possible explanations for these anomalies, which could help facilitate measures to mitigate CAW both at the state and national level in India.

3 | Literacy and Women's Empowerment in India

There have been consistent efforts and initiatives towards nation-building through education in India. The introduction of mid-day meal schemes played a vital role in reducing school drop-out rates. The Sarva Shiksha Abhiyan (SSA) scheme of the government (Sarva Shiksha Abhiyan Scheme 2001) and the Right to Education (RTE) Act (Right to Education 2009) are landmark policies towards providing free and compulsory education to children aged 6–14 years. The National Literacy Mission (NLM), National Literacy Mission Authority (1988), implements adult education programs to teach the adult population essential reading, writing and numeracy skills. The Digital India Initiative (Digital India 2015) has tried to leverage technology to enhance education through the digital literacy of students and teachers.

India exemplifies wide diversity in size, demography and developmental indices among its 28 states and eight union territories (as of 2023). Overall, literacy is considerably different in these states. Table 1 indicates the changes in LRs in the major states and union territories across five decades (1981–2022). Although LRs in all states have increased since 1981, there remains a wide discrepancy, especially in the larger states, ranging from a high of 95.12% in Kerala to a low of 72.10% in Andhra Pradesh in 2022.

This variation within states can be attributed to diverse socioeconomic factors, cultural disparities and levels of access to education. Those states indicating higher LRs tend to prioritise education and have better infrastructure, leading to improved educational outcomes. Conversely, those states with lower LRs struggle with broad challenges such as poverty, gender inequality and inadequate educational facilities. Bridging this gap through targeted interventions, equitable resource allocation and community engagement is pivotal to guarantee equal opportunities for all.

Realising that women's education can be a powerful catalyst for social transformation, the government has initiated many programmes. The Beti Bachao, Beti Padhao (BBBP 2015) (Save the Girl, Educate the Girl) campaign aimed to address the skewed gender gap in education. The Mahila Samakhya Programme (MSP 1988) focuses on women's empowerment and education in rural areas. This has resulted in a considerable increase in the LRs in India. Figure 1 indicates India's consistent increase in overall LRs and FLRs over seven decades. Although the total LR was 18.32% in 1951, it increased to 79.4% by 2022. The FLR has also increased from 8.86 in 1951 to 74.2% in 2022, although there is a considerable gap in male-FLRs.

4 | The Enduring Curse of CAW

Economic disparities, technological advancements, diminishing global resources and social unrest have contributed to the proliferation of crime and evolving criminal dynamics across the world. In India, rapid urbanisation, socio-economic disparities, communal tensions, political discord and digital innovations have all contributed to a wide range of crimes ranging from petty theft, robbery and violent crimes to complex financial fraud and

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Literacy rates by percentage in Indian states by decades from 1981 to 2022.

States	1981	1991	2001	2011	2022
Andhra Pradesh	25.54	44.09	60.47	67.66	72.10
Arunachal Pradesh	35.66	41.59	54.34	66.95	84.03
Assam	NA	52.89	63.25	73.18	77.46
Bihar	32.03	37.49	47.00	63.82	74.30
Chhattisgarh	NA	42.91	64.66	71.04	78.04
Goa	65.71	75.51	82.01	87.4	95.04
Gujarat	52.21	61.29	69.97	79.31	85.10
Haryana	43.85	55.85	67.91	76.64	82.83
Himachal Pradesh	51.17	63.86	76.48	83.78	88.15
Jammu and Kashmir	32.68	NA	55.52	68.74	78.16
Jharkhand	NA	NA	53.56	67.63	79.39
Karnataka	46.20	56.04	66.64	75.6	80.72
Kerala	81.56	89.81	90.92	93.91	95.12
Madhya Pradesh	34.22	44.67	63.74	70.63	76.16
Maharashtra	55.83	64.87	76.88	82.91	86.77
Manipur	49.61	59.89	70.53	79.85	90.64
Meghalaya	42.02	49.10	62.56	75.48	95.05
Mizoram	74.26	82.27	88.80	91.58	99.43
Nagaland	50.20	61.65	66.59	80.11	95.55
Orissa	40.96	49.09	63.08	73.45	79.30
Punjab	48.12	58.51	69.65	76.68	83.25
Rajasthan	30.09	38.55	60.41	67.06	73.21
Sikkim	41.57	56.94	68.81	82.2	87.08
Tamil Nadu	54.38	62.66	73.45	80.33	86.02
Tripura	50.10	60.44	73.19	87.75	93.24
Uttar Pradesh	33.33	40.71	56.27	69.72	76.59
Uttarakhand	NA	57.75	71.62	79.63	86.00
West Bengal	48.64	57.70	68.64	77.08	82.84
A and N Islands	63.16	73.02	81.30	86.27	92.70
Chandigarh	74.81	77.81	81.94	86.43	92.65
D and N Haveli	32.70	40.71	60.03	77.65	86.99
Daman and Diu	59.91	71.20	81.09	87.07	86.99
Delhi	71.93	75.29	81.67	86.34	88.01
Lakshadweep	68.42	81.78	86.66	92.28	98.20
Puducherry	65.14	74.74	81.24	86.55	91.28
India	**43.56**	**52.21**	**64.84**	**74.04**	**74.04**

Source: Census Digital Library (2023), Women & Men in India—A Statistical Compilation of Gender-Related Indicators in India 2021 (Women & Men in India 2022), National Sample Survey Office (NSSO 2023).

cybercrimes. Figure 2 indicates India's gradual increase in overall crime rates between 1948 and 2019. Of note is the phenomenal jump in overall crime rates after 1990.

In India, women grapple with increased personal violations that stem from an all-pervasive mindset of patriarchy and traditional gender roles. This perpetuates the stereotypical notion that women are inferior, subordinate and therefore subject to male dominance and control throughout life. Despite a plethora of constitutional guarantees and legislations that champion gender equality, for example, the Criminal Law (Amendment) Act of 2013 (Government of India 2013), which strengthened laws relating to sexual offences, offenders continue to commit CAW with impunity. Figure 3 indicates the increase in CAW over three

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

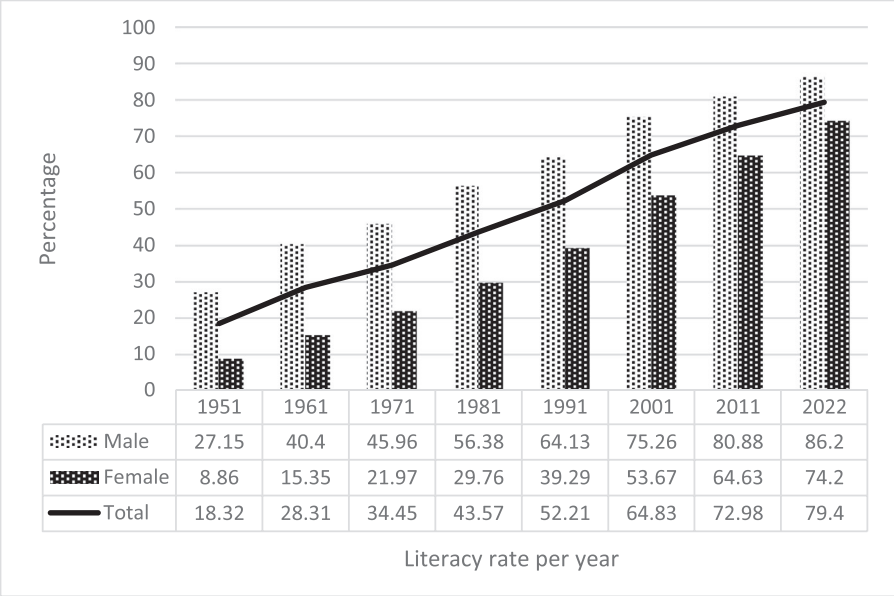

FIGURE 1 | Male and female literacy rates in India by decades from 1951 to 2022. *Source:* Census Digital Library (2023), Women & Men in India (Women & Men in India 2022), National Sample Survey Office (NSSO 2023).

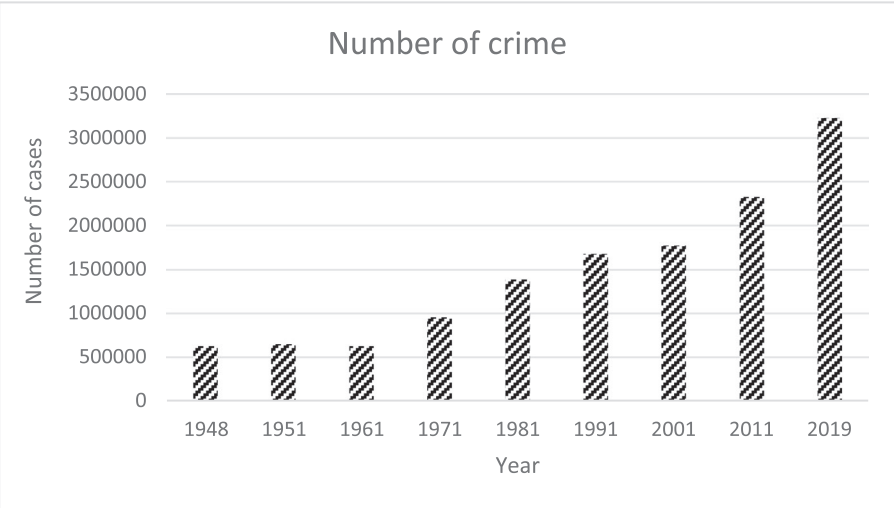

FIGURE 2 | Overall crimes in India (Indian Penal Code crimes [IPC] and Special and Local Law crime [SLL]) from 1948 to 2019. *Source: Crime in India* reports from 1948 to 2019 (Crime in India 2023).

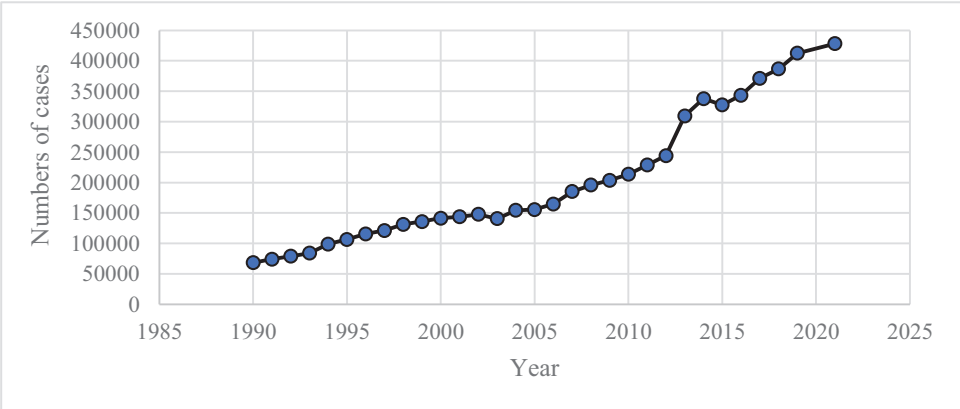

FIGURE 3 | Crimes against women in India (IPC and SLL) from 1990 to 2021. *Source: Crime in India*, reports from 1990 to 2021 (Crime in India 2023). [Colour figure can be viewed at wileyonlinelibrary.com]

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Categories and statistics of specific crimes against women (CAW) (Indian Penal Code crimes [IPC] and Special and Local Law crime [SLL]) from 1991 to 2021.

Specific CAW	1991	2001	2011	2021
Importation of girls from foreign countries	NA	114	80	2
Rape	9793	16,075	2420	31,677
Abduction and kidnapping	12,300	14,646	35,565	101,707
Dowry death	5157	6851	8618	6757
Cruelty and torture by husband and in-laws	15,949	49,170	99,135	123,442
Molestation-Assault with intent to outrage modesty	20,611	34,124	42,968	89,200
Acid attack	NA	NA	NA	176
Sexual harassment-insult to modesty	10,283	9746	8570	17,539
Sexual harassment at work or office premises	NA	NA	NA	418
Sexual harassment in public transport systems	NA	NA	NA	357
Assault or criminal force with intent to disrobe	NA	NA	NA	11,102
Kidnapping and abduction to compel for marriage	NA	NA	NA	28,012
Suppression of Immoral Traffic in Women and Girls Act (1956)	14,639	8796	2436	1678
Dowry Prohibition Act (1961)	1841	3222	6619	13,568
Indecent Representation of Women Act (1986)	NA	1052	453	28
Commission of Sati (Prevention) Act (1987)	17	0	0	0
Protection of women from domestic violence Act (2005)	NA	NA	NA	507

Abbreviation: NA, not available.
Source: Crime in India—reports from year 1991–2021 (Crime in India 2023).

decades from 1990 to 2021. It must be noted that since 2005, these crimes have increased over three times.

In India, violations against women include domestic violence, such as physical, emotional and economic abuse inflicted by husbands and family members; sexual assault, including marital rape; dowry-related harassment, violence and death at the hands of husbands and in-laws; human trafficking (for sexual exploitation, forced labour and marriage), revenge instigated acid attacks, kidnapping for ransom or revenge, child marriage, cybercrimes (cyberbullying, online harassment and blackmail through non-consensual sharing of intimate images), female foeticide and infanticide, honour killings (by families and community) for transgressing societal marriage norms of caste and class, forced arranged marriages as well as stalking (online and offline). As seen, some of these crimes specifically relate to Indian contexts.

The socio-cultural psyche in India discourages reporting of sexual CAW due to fear of societal ostracism, shame, rebuke and blame. Poverty and illiteracy, especially in rural India, make women vulnerable to sexual exploitation and violence. Table 2 shows the different CAWs in India and how they have increased over four decades. In the case of rape, abduction and kidnapping, domestic abuse by husbands and in-laws and all specific cases of molestation and sexual harassment, there has been a phenomenal increase over the four decades.

A graph comparing overall crime rates (Figure 2) with CAW in India (Figure 3) is depicted in Figure 4. As noted, even as the rates of overall crimes have increased, CAW also indicates a steady increase.

Further, a graph correlating state-wise female literacy and state-wise CAW in India is depicted in Figure 5, and a graph comparing female literacy and CAW in the Union territories is depicted in Figure 6. These indicate that in most states with low female literacy, such as Andhra Pradesh, Bihar, Madhya Pradesh and Uttar Pradesh, the rate of CAW is very high, which supports our understanding that female literacy is correlated with CAW.

However, some anomalies are noticed in a few states with higher CAW despite higher female literacy. In Maharashtra, where female literacy is 75.9%, CAW remains marginally high; in West Bengal, where female literacy is 70.5%, CAW remains very high. The most puzzling anomaly is in Kerala, where CAW remains high despite attaining the highest female literacy in India at 92.1% in 2022. In the case of most union territories (Figure 6), the LRs were high, and CAW was low. However, the exception was Delhi, India's capital, where CAW remains the highest in India despite female literacy of 80%.

5 | Methodology

The objectives of this study were to analyse the trajectories of female literacy and CAW in India to determine if there is a positive correlation between the two, and whether female literacy plays a role in mitigating CAW. Statistical data were collected from the Office of the Registrar General & Census Commissioner—Census of India reports (Census Digital Library 2023), from Ministry of Statistics and Programme Implementation—Women & Men in India reports (Women & Men in India 2023), and from National Crime Records Bureau (NCRB)—Crime in India (2023) from 1990

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

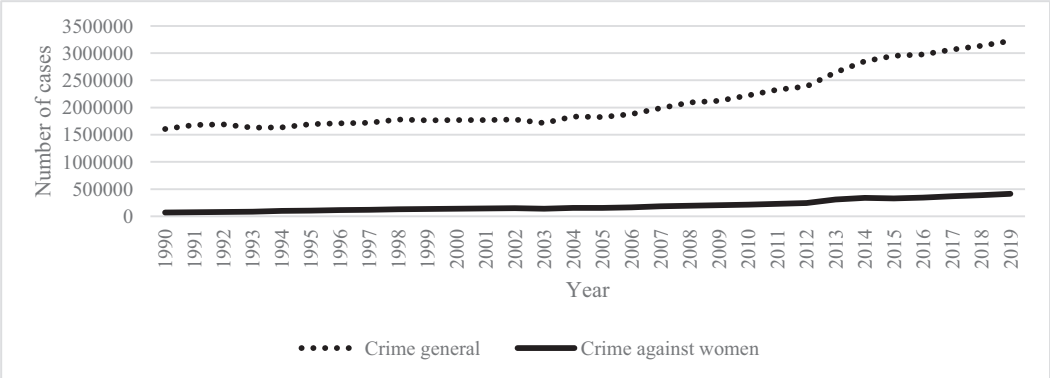

FIGURE 4 | Comparative trajectory of total crimes and CAW in India from 1990 to 2021. *Source: Crime in India*—reports from 1990 to 2021 (Crime in India 2023).

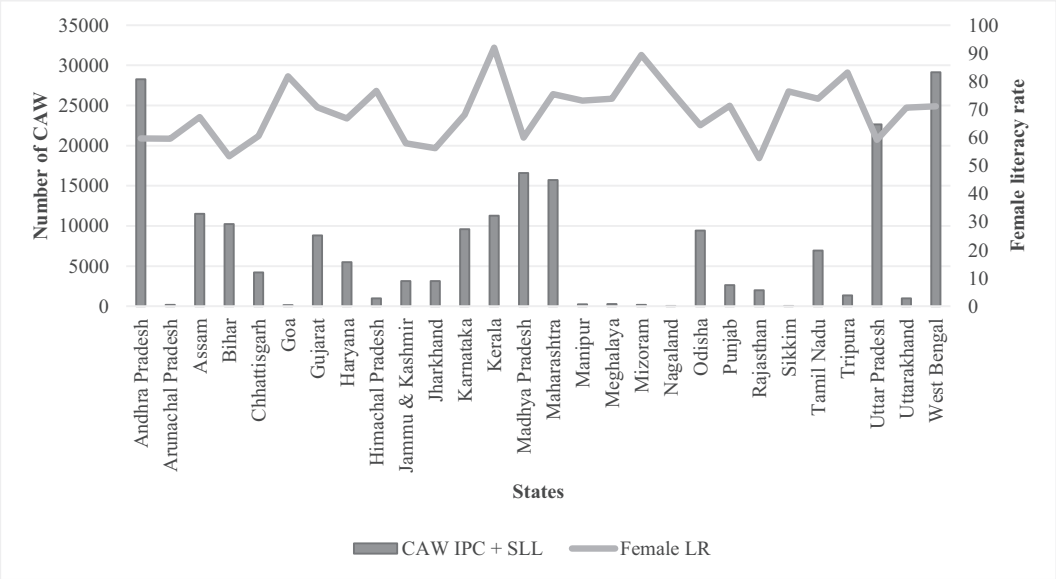

FIGURE 5 | Female literacy and CAW in India by states. CAW, crimes against women; IPC, Indian Penal Code crimes; LR, literacy rate; SLL, Special and Local Law crime. *Source:* Census Digital Library (2023), Crime in India (2023), *National Sample Survey Office (NSSO* 2023).

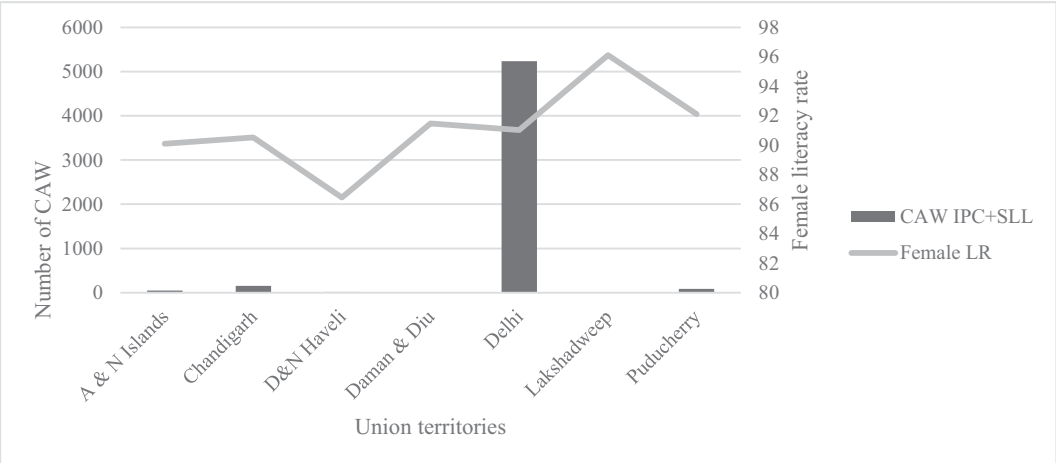

FIGURE 6 | Female literacy and CAW in India by union territories. CAW, crimes against women; IPC, Indian Penal Code crimes; LR, literacy rate; SLL, Special and Local Law crime. *Source:* Census Digital Library (2023), Crime in India (2023), *National Sample Survey Office (NSSO* 2023).

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Correlation of total crimes against women (CAW) (Indian Penal Code crimes [IPC] and Special and Local Law crime [SLL]) and total literacy rate (LR).

		Total CAW	Total LR
Total CAW	Pearson correlation	1	0.937
	Sig. (2-tailed)		0.063
	N	32	4
Total LR	Pearson correlation	0.937	1
	Sig. (2-tailed)	0.063	
	N	4	4

to 2021. These quantitative data were tabulated and analysed using the statistical software SPSS, which showed considerable correlation.

To test trajectories of total LR, FLR and CAW, the authors used a dataset of variables: LR (percentage of male and female aged 7 years and above who can read and write with understanding), FLR (percentage of females aged 7 years and above who can read and write with understanding) and total CAW (IPC—Indian Penal Code crimes and SLL—Special and Local Law crime).

5.1 | Research Model

Regression analysis (Field 2009) is used to predict the values of a dependent variable given the values of one or more independent variables by calculating a regression equation. For example, the following research model was designed to find the impact of LR on CAW:

$$\mathrm{CAW} = \beta_0 + \beta_1 \mathrm{LR} + \mathrm{e_i}$$

where CAW is the dependent variable that is to be predicted. β_0 is the regression constant, representing a value of CAW if there is no LR (or FLR). β_1 is the beta coefficient. LR is the value of total LR (or FLR). e_i represents error terms for all other factors that account for variance in the dependent variable.

6 | Results

To validate our research objectives, we analysed the correlation of LR and FLR against total CAW to prove the following two research problems with supporting hypotheses.

Null hypothesis (H0). *Total LR has no significant impact on total CAW.*

Alternative hypothesis H1. *Total LR has a significant impact on total CAW. A bivariant correlation was used to understand the correlation between two variables (total CAW and total LR) (Table 3). The probability is 0.063, which is less than 0.1. The relationship is significant at 10%. The correlation between total CAW and total LR is 0.937. Evans (1996) suggests that the correlation is very strong when the absolute value of* r *is between 0.8 and 1. Our findings conclude that total CAW and total LR have a robust correlation.*

The hypothesis tests if total LR carries a significant impact on total CAW. The dependent variable total CAW was regressed on the independent variable LR to test hypothesis H1 (Table 4). Total LR significantly predicted total CAW, F (1,2) = 14,361, $p < 0.1$, which indicates that the LR can play a significant role in CAW ($b = 0.937$, $p < 0.1$). These results show a significant impact of LR on CAW. Additionally, $R^2 = 0.878$ indicates that LR can account for an 87.8% change in CAW. (This means that LR influences 87% of the change in CAW.) The hypothesis validation confirms the strength of these predictors in explaining the variations in the dependent variable (CAW), highlighting the importance of educational initiatives and addressing CAW in India. The results indicate that the alternative hypothesis (H1) is supported, as total LR significantly predicts total CAW.

Having established the significant impact of LR on CAW in India, we sought to check our second research problem. The second research problem was to find the correlation between FLR and total CAW. Null hypothesis (H0): FLR has no significant impact on total CAW. Alternative hypothesis H2: FLR has a significant impact on total CAW. A bivariant correlation was used to understand the correlation between two variables (total CAW and FLR) (Table 5). As the probability is 0.058, which is less than 0.1, the relationship is significant at 10%. The correlation between total CAW and FLR is 0.942. Our findings conclude that total CAW and FLR have a very strong correlation.

The hypothesis tests if FLR carries a significant impact on total CAW. The dependent variable CAW was regressed on the independent variable FLR to test hypothesis H2 (Table 6). FLR significantly predicted CAW, F (1,2) = 15,865, $p < 0.1$, which indicates that FLR can play a significant role in CAW ($b = 0.942$, $p < 0.1$). These results show a significant impact of total FLR on total CAW. Additionally, the $R^2 = 0.888$ indicates that FLR can account for an 88.8% change in CAW. (This means that FLR influences 88.8% of the change in CAW). The p value of 0.058 is slightly above the conventional significance level of 0.05. However, depending on the study's threshold for significance (e.g., $p < 0.1$), this p value could still indicate a statistically significant relationship between FLR and CAW. The model's predictors (FLR and CAW) are statistically significant. The hypothesis validation confirms the robustness of these predictors in explaining the variations in the dependent variable (CAW), highlighting the importance of educational initiatives for females and addressing CAW in India. The results indicate that the alternative hypothesis (H1) is supported, as total FLR significantly predicts total CAW.

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Impact of total literacy rate (LR) on total crimes against women (CAW) in India.

Hypothesis	Regression weights	Beta coefficient	R^2	F	p value	Hypothesis validated
H1	LR → CAW	0.937	0.878	14,361	0.063	YES

TABLE 5 | Correlation of total female literacy rate (FLR) to total crimes against women (CAW).

		Total FLR	Total CAW
Total FLR	Pearson correlation	1	0.942
	Sig. (2-tailed)		0.058
	N	4	4
Total CAW	Pearson correlation	0.942	1
	Sig. (2-tailed)	0.058	
	N	4	32

TABLE 6 | Impact of total female literacy rate (FLR) and total crimes against women (CAW) in India.

Hypothesis	Regression weights	Beta coefficient	R^2	F	p value	Hypothesis validated
H2	FLR → CAW	0.942	0.888	15,865	0.058	YES

The table suggests that there is a strong, statistically significant relationship between FLR and CAW, with FLR explaining a large portion of the variance in CAW. The hypothesis H2 is validated, meaning that the data support the conclusion that FLR influences CAW significantly.

7 | Discussion

Despite the overall increase in literacy and female literacy in India, there is a significant increase in CAW, which validates this study to explore the links between female literacy and CAW. Regarding our first research question, determining the correlation between total LRs and CAW in India, we found through hypothesis 1 that there is a significant correlation and impact of total LR on total CAW in India, as seen in Tables 2 and 3.

Our second research question was determining the correlation between the FLR and CAW in India. Here, we tested hypothesis 2, which showed a significant correlation and impact of female literacy on total CAW (Tables 4 and 5). This supports our inference that in most states where there is increased literacy, CAW is relatively lower, which suggests that women are better empowered to fight discrimination, harassment and violence through legal recourse, report offences against them and seek remedies, thereby reducing their vulnerability to such crimes.

Our third research question was to find the trajectories of FLRs and CAW in the Indian states and union territories. Our comparative analysis of data revealed that in most of India's states and union territories, there is a significant correlation between higher female literacy and lower CAW (Figures 5 and 6).

Answering our final research question on the significance of the correlation between female literacy and CAW and whether there are any anomalies to the patterns, we noticed that although there is considerable significance of female literacy on CAW, there are anomalies in some states, where even though literacy levels were high, CAW also indicated an increasing trend. Predominant anomalies are observed in three key states, which are discussed below:

Kerala, with the highest FLR in India (92.1% in 2022), is often lauded for the 'Kerala state developmental model' (Franke and Chasin 1998; Kurien 1995; Parayil 1996) that is indicated by high literacy levels, health indicators, life expectancy as well as reduced maternal mortality and child mortality rates. However, despite such hype, Kerala recorded a CAW rate of 73.3% in 2021 (NCRB 2021), indicating that women in Kerala remain powerlessness despite securing high literacy (Mukhopadhyay 2011) and suggesting that educational attainment alone cannot determine women's empowerment (J. Simister 2011). Historically, many communities in Kerala, such as the *Nairs*, followed the matrilineal system, and Kerala was a frontline in the fight for social reform and gender equality. However, achievements in many developmental matrices notwithstanding, the women of Kerala suffer discrimination and sexual harassment at work (which might account for the low work participation of women), dowry demands, domestic violence and high suicide rates (Mitra and Singh 2007). The recent report by the government-appointed Justice Hema Committee unveiled rampant sexual abuses and workplace violations against women, including casting couch allegations and sexual harassment in the Kerala film industry, prompting another #metoo movement. The higher statistics of CAW suggest the need to reevaluate the famed Kerala model, as it negates the assumption that literacy and crime rates are negatively correlated (Joseph and Mathew 2019). Panda (2004) suggests that considering the rising cases of alcoholism in Kerala, there might be a strong association between excessive consumption of alcohol and experience of physical and psychological violence.

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Delhi is the national capital of India, with an FLR of 80.0% in 2022. However, it is also the state with the highest cases of CAW, with a rate of 147.6% in 2021 (NCRB 2021), highlighting the need for comprehensive societal changes. One of the most distressing categories of CAW in Delhi is domestic violence (K. Sharma et al. 2019), as well as sexual assault, rape and molestation in public spaces (Govinda 2020). The *Nirbhaya* gang rape and murder case of 2012 in Delhi brought the issue of gender-based violence into public discourse, highlighting the need for a safe environment for women. It sparked media debates on the lack of effective law enforcement, poor street lighting, inadequate transportation and insufficient public awareness campaigns. These factors contribute to an ethos where perpetrators feel emboldened. Delhi's workplace environment also poses challenges for women, with instances of sexual harassment and discrimination reported across various industries. This suggests that women continue to live in fear (Iyer 2019), and the efforts to mitigate CAW in Delhi (Atluri 2013) need to look beyond empowering through education and literacy (Nieder, Muck, and Kärtner 2019; Reddy and Acharya 2017).

West Bengal had an FLR of 70.5% in 2022, which is above the national average, yet CAW in West Bengal has been on the rise. The rate of CAW in 2021 was 74.6% (NCRB 2021) and encompasses dowry (Moulick et al. 2022), sexual assault and rape (Chatterjee, Choudhury, and Bandyopadhyay 2022), acid attacks, human trafficking (P. Biswas and Chatterjee 2021), child marriage (A. Roy and Chouhan 2021) and domestic abuse and harassment (C. S. Biswas, Pal, and Bharati 2022). West Bengal is a state where women are historically and traditionally immortalised as Goddess *Durga* or *Shakti*, capable of immense power and courage. However, cases like the *Abhaya* rape and murder case of a resident doctor inside a hospital in Kolkata in August 2024 point to systemic failures that impede the safety of women. Therefore, despite having a good level of literacy, other factors like unemployment, poverty, mental health, lack of financial independence, lack of legal awareness and a social culture of hyper masculinity and patriarchy (A. Roy 2019; S. Roy, Bose, and Chowdhury 2023) contribute to CAW in West Bengal.

The links between literacy and CAW in India are complex and multifaceted. Literacy empowers women with knowledge, awareness and skills that contribute to reducing their vulnerability to various forms of violence. It challenges societal norms, promotes economic independence and fosters a generation of individuals who are more likely to reject violence and discrimination. Therefore, investing in women's education is crucial to creating a safer and more equitable society. As India strives to address CAW, promoting total literacy and female literacy is a powerful tool for change and progress. However, it is essential to recognise that LRs alone may not eradicate CAW in India. There may also be several other factors that might facilitate the reduction of CAW in India, such as restriction or prohibition of alcohol (Luca, Owens, and Sharma 2015), improved women's eligibility for inheritance (Amaral 2017), more balanced sex ratio (Amaral and Bhalotra 2017; Kaur et al. 2016; Verma, Qureshi, and Kim 2017) and higher female workforce participation (Agarwal Goel 2021; Maity and Roy 2021), among others.

8 | Conclusion

For any society, education and enhanced literacy are fundamental for the empowerment of citizens, enabling them to critique injustice and transform societal evils. Education is a potent tool for empowering women, fostering gender equality and driving societal progress. Educated women are more likely to make better health and family planning choices, resulting in improved maternal and child health outcomes. Women's education can also contribute to the country's economic growth by increasing the female workforce's productivity and economic participation. Further education challenges traditional gender roles, leading to more equitable opportunities, less discrimination and better respect for womanhood within society. Educated women are more likely to invest in their children's education, ending the intergenerational poverty cycle.

The development of female literacy in India has been due to varied initiatives by the government, civil society and private institutions. However, gender bias, discrimination and inequality are sustained by traditional socio-cultural norms and practices of patriarchy. Many women remain constrained into domestic roles within private household spaces and as caregivers. The practice of early marriage has resulted in high school dropout rates so that women can assume domestic duties within households. Poverty and lack of financial resources present barriers to female access to education, as families facing financial hardships tend to prioritise the education of male children as 'potential' breadwinners of the family. Concerns over women's safety, especially if they have to travel long distances to schools or study in co-educational schools, may deter parents from educating the women. The lack of a gender-sensitive curriculum might make women students uncomfortable or embarrassed by gendered comments and behaviour in co-educational settings. Further, poverty, social disparities, cultural norms, erratic implementation of laws and the social stigma against victims of gender violence continue to propel CAW.

This article delves into female literacy as a mitigating factor in decreasing CAW in India. It is based on the premise that educated women would have better access to career and leadership, be more aware of their legal rights, and are likely to reduce their vulnerability to crimes. The main objectives of this study were to analyse the trajectories of female literacy and CAW at the national and state levels in India to determine if there is a positive correlation between the two. Our quantitative data analysis from governmental databases using the statistical software SPSS indicated a considerable positive correlation between female literacy and CAW. However, certain anomalies were noticed in a few states where, despite higher female literacy, CAW against women was also on the rise. Noticing these variations in our comparative correlations (Figures 5 and 6), we used bivariate regression analysis to derive answers to our research questions and validate whether higher female literacy impacts CAW.

The research does point to the strong link between FLRs and a decrease in CAW in most states in India, which indicates that literacy is a crucial tool in educating and empowering women, challenging stereotypical gender norms and promoting

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

gender equality. However, to effectively combat CAW, a more comprehensive approach is needed. As some anomalies in our study suggest, higher levels of female literacy alone do not automatically translate to reduced CAW. Addressing socioeconomic factors such as eradicating poverty and unemployment, increasing women in the workforce, reducing alcoholism, increasing legal literacy and updating and ensuring enforcement of laws can all help bring about an ethos favouring women. In July 2024, the Indian government has replaced existing criminal laws with three new criminal laws: the Bharatiya Nyaya Sanhita (BNS), the Bhartiya Nagarik Suraksha Sanhita (BNSS) and the Bhartiya Sakshya Adhiniyam (BSA) which demonstrate an important step in revamping the 150 years of the colonial criminal justice system in India. However, these have been criticised for ignoring the long-standing recommendation to separate the investigation wing from the law and order wing for better policing, for the absence of pre-legislative consultation, for lack of significant reform, particularly in CAW, and for lack of clarity in implementation (Moti 2024). New laws notwithstanding, the primary need of the hour is transforming mindsets against gender stereotypes, and bias is a task for all stakeholders and requires concentrated and consistent efforts.

This article was constrained in securing updated data from official Census reports for LRs as these are available only until 2011 since the next census has been delayed and conducted on a date in India, largely attributed to the COVID-19 pandemic and the national elections in 2024. However, the authors have supplemented more recent statistics from official sample surveys wherever available. For crime statistics, the authors relied on NCRB data. However, since most cases of CAW in India remain unreported, it can be assumed that actual statistics are much higher than officially stated in reports. These factors present limitations to this article. Further, the authors focused on the links between female literacy and CAW. In contrast, the role of female workplace participation and the efficacy of legislation and law enforcement were beyond the scope of this article. The authors suggest that this could be the focus of future research.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article.

Peer Review

The peer review history for this article is available at: https://publons.com/publon/10.1111/issj.12550.

References

Agarwal Goel, P. 2021. "Effects of Female Workforce Participation on Crime: An Analysis From the Districts of India." *Journal of Public Affairs* 21, no. 1: e2165. https://doi.org/10.1002/pa.2165.

Amaral, S. 2017. Do Improved Property Rights Decrease Violence Against Women in India?. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2504579.

Amaral, S., and S. Bhalotra. 2017. Population Sex Ratios and Violence Against Women: The Long-Run Effects of Sex Selection in India (SSRN Scholarly Paper 3055794). Rochester, NY: SSRN. https://doi.org/10.2139/ssrn.3055794.

Atluri, T. 2013. "The Young and the Restless. Gender, 'Youth', and the Delhi Gang Rape Case of 2012." *Sikh Formations* 9, no. 3: 361–379.

Beti Bachao Beti Padhao: Caring for the Girl Child. 2015. Prime Minister, Government of India. https://www.pmindia.gov.in/en/government_tr_rec/beti-bachao-beti-padhao-caring-for-the-girl-child/.

Bisika, T., P. Ntata, and S. Konyani. 2009. "Gender-Violence and Education in Malawi: A Study of Violence Against Girls as an Obstruction to Universal Primary School Education." *Journal of Gender Studies* 18, no. 3: 287–294. https://doi.org/10.1080/09589230903057183.

Biswas, C. S., M. Pal, and P. Bharati. 2022. "Violence Against Women in West Bengal: Its Extent and Causes." *Journal of Gender-Based Violence* 6, no. 1: 149–171. https://doi.org/10.1332/239868021x16172761779557.

Biswas, P., and N. D. Chatterjee. 2021. "Factors Persuading Acid Attacks, a Modern Revenge Strategy Against Women: A Study in West Bengal, India." *Indian Journal of Geography and Environment* 17, no. 18: 106–117.

Census of India. 2023. *Office of the Registrar General & Census Commissioner, India, Ministry of Home Affairs, Government of India*. New Delhi: Census of India. https://censusindia.gov.in/nada/index.php/catalog/central/?page=1&sk=literacy&sort_by=popularity&sort_order=desc&ps=15.

Chatterjee, S., S. Choudhury, and A. R. Bandyopadhyay. 2022. "Recapitulation of Rape: A Spatial and Temporal Analysis of Rape in West Bengal, India." *Antrocom: Online Journal of Anthropology* 18, no. 2: 503–516.

Crime in India. 2023. *National Crime Records Bureau, Ministry of Home Affairs, Government of India*. New Delhi: National Crime Records Bureau. https://www.ncrb.gov.in/crime-in-india.html.

Das, T., and D. T. Basu Roy. 2020. "More Than Individual Factors; Is There any Contextual Effect of Unemployment, Poverty and Literacy on the Domestic Spousal Violence Against Women? A Multilevel Analysis on Indian Context." *SSM—Population Health* 12: 100691. https://doi.org/10.1016/j.ssmph.2020.100691.

Deyessa, N., Y. Berhane, M. Ellsberg, M. Emmelin, G. Kullgren, and U. Högberg. 2010. "Violence Against Women in Relation to Literacy and Area of Residence in Ethiopia." *Global Health Action* 3, no. 1: 2070.

Digital India. 2015. India AI. https://indiaai.gov.in/missions/digital-india.

Drèze, J., and R. Khera. 2000. "Crime, Gender, and Society in India: Insights From Homicide Data." *Population and Development Review* 26, no. 2: 335–352. https://doi.org/10.1111/j.1728-4457.2000.00335.x.

Erten, B., and P. Keskin. 2018. "For Better or for Worse?: Education and the Prevalence of Domestic Violence in Turkey." *American Economic Journal: Applied Economics* 10, no. 1: 64–105. https://doi.org/10.1257/app.20160278.

Evans, J. D. 1996. *Straightforward Statistics for the Behavioral Sciences*. Pacific Grove, CA: Thomson Brooks/Cole Publishing Co.

Field, A. 2009. *Discovering Statistics Using SPSS*. Thousand Oaks, CA: SAGE Publications Ltd.

Fleming, P. M. 2000. "Three Decades of Educational Progress (and Continuing Barriers) for Women and Girls." *Equity & Excellence in Education* 33, no. 1: 74–79. https://doi.org/10.1080/1066568000330111.

Franke, R., and B. Chasin. 1998. "The Kerala Model of Development a Debate (Part 1). Kerala: A Valid Alternative to the New World Order." *Bulletin of Concerned Asian Scholars* 30: 25–28.

Gallaway, J. H., and A. Bernasek. 2004. "Literacy and Women's Empowerment in Indonesia: Implications for Policy." *Journal of Economic Issues* 38, no. 2: 519–525. https://doi.org/10.1080/00213624.2004.11506713.

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gill, A. K., and H. Mason-Bish. 2013. "Addressing Violence Against Women as a Form of Hate Crime: Limitations and Possibilities." *Feminist Review* 105, no. 1: 1–20. https://doi.org/10.1057/fr.2013.17.

Government of India. 2013. The Criminal Law (Amendment) Act. Ministry of Law and Justice. https://www.iitk.ac.in/wc/data/TheCriminalLaw.pdf.

Govinda, R. 2020. "From the Taxi Drivers' Rear-View Mirror: Masculinity, Marginality and Sexual Violence in India's Capital City, Delhi." *Gender, Place & Culture* 27, no. 1: 69–85. https://doi.org/10.1080/0966369X.2019.1567469.

Hazra, D. 2020. "What Does (and Does not) Affect Crime in India?" *International Journal of Social Economics* 47, no. 4: 503–521. https://doi.org/10.1108/IJSE-03-2019-0206.

Hill, M. A., and E. King. 1995. "Women's Education and Economic Well-Being." *Feminist Economics* 1, no. 2: 21–46. https://doi.org/10.1080/714042230.

Himabindu, B. L., R. Arora, and N. S. Prashanth. 2014. "Whose Problem Is It Anyway? Crimes Against Women in India." *Global Health Action* 7, no. 1: 23718. https://doi.org/10.3402/gha.v7.23718.

Htun, M., and F. R. Jensenius. 2020. "Fighting Violence Against Women: Laws, Norms & Challenges Ahead." *Daedalus* 149, no. 1: 144–159. https://doi.org/10.1162/daed_a_01779.

Iyer, P. 2019. "'Due to all This Fear, We're Getting Less Freedom': Young People's Understandings of Gender and Sexual Violence in New Delhi, India." *Gender and Education* 31, no. 2: 222–239. https://doi.org/10.1080/09540253.2017.1306029.

Jahromi, M. K., S. Jamali, A. R. Koshkaki, and S. Javadpour. 2016. "Prevalence and Risk Factors of Domestic Violence Against Women by Their Husbands in Iran." *Global Journal of Health Science* 8, no. 5: 175–183. https://doi.org/10.5539/gjhs.v8n5p175.

Joseph, J., and J. Mathew. 2019. "The Relationship Between Literacy Rate and Crime Rate: An Analysis With Reference to Kerala." *International Journal of Research and Analytical Reviews* 6, no. 1: 202–205.

Kabeer, N. 1999. "Resources, Agency, Achievements: Reflections on the Measurement of Women's Empowerment." *Development and Change* 30, no. 3: 435–464. https://doi.org/10.1111/1467-7660.00125.

Kabeer, N. 2019. "Women's Empowerment and the Question of Choice." *Journal of International Affairs* 72, no. 2: 209–214.

Kaur, R., S. S. Bhalla, M. K. Agarwal, and P. Ramakrishnan. 2016. *Sex Ratio Imbalances and Crime Rates*. New York, NY: United Nations Population Fund.

Kumar, S., and S. R. Kuncharam. 2020. "Determinants of Women Empowerment Responsible for Reducing Crime against Women in India." *Violence and Gender* 7, no. 4: 182–187. https://doi.org/10.1089/vio.2020.0023.

Kurien, J. 1995. "The Kerala Model: Its Central Tendency and the Outlier." *Social Scientist* 23, no. 1/3: 70–90. https://doi.org/10.2307/3517892.

Lawoko, S., K. Dalal, L. Jiayou, and B. Jansson. 2007. "Social Inequalities in Intimate Partner Violence: A Study of Women in Kenya." *Violence and Victims* 22, no. 6: 773–784. https://doi.org/10.1891/088667007782793101.

Lolayekar, A. P., S. Desouza, and P. Mukhopadhyay. 2022. "Crimes against Women in India: A District-Level Analysis (1991–2011)." *Journal of Interpersonal Violence* 37, no. 9–10: NP7289–NP7314. https://doi.org/10.1177/0886260520967147.

Luca, D. L., E. Owens, and G. Sharma. 2015. "Can Alcohol Prohibition Reduce Violence Against Women?" *American Economic Review* 105, no. 5: 625–629. https://doi.org/10.1257/aer.p20151120.

Mahila Samakhya Programme. 1988. Governmentof India, Ministry of Education, Madras. https://www.education.gov.in/mahila-samakhya-programme.

Maity, S., and S. Roy. 2021. "Analysis of Growth and Identifications of the Determinants of Crime Against Women: Insight From India." *Journal of International Women's Studies* 22, no. 1: 293–311.

Marium, S. 2014. "Women's Level of Education and Its Effect on Domestic Violence in Rural Bangladesh." *IOSR Journal of Humanities and Social Science* 19, no. 5: 40–45. https://doi.org/10.9790/0837-19534045.

Mitra, A., and P. Singh. 2007. "Human Capital Attainment and Gender Empowerment: The Kerala Paradox*." *Social Science Quarterly* 88, no. 5: 1227–1242. https://doi.org/10.1111/j.1540-6237.2007.00500.x.

Morrisroe, J. 2014. *Literacy Changes Lives 2014: A New Perspective on Health, Employment and Crime*. London: National Literacy Trust.

Moti, S. 2024. Passing of new criminal law bills: Amissed opportunity for reform? *Bar and Bench*. https://www.barandbench.com/columns/passing-of-criminal-law-bills-in-the-parliament-a-missed-opportunity-for-reform.

Moulick, J., M. Dasgupta, S. G. N. Ghatak, N. Thakur, and P. K. Sarkar. 2022. "Dowry Demand in West Bengal: A Socio-Legal and Historical Analysis." *Liberal Studies* 7: 110.

Mukhopadhyay, S. (Ed.). 2011. *The Enigma of the Kerala Woman: A Failed Promise of Literacy*. Hyderabad: Orient Blackswan.

National Crime Records Bureau (NCRB). 2021. Crime in India 2020, Statistics. New Delhi: National Crime Records Bureau, Ministry of Home Affairs, Government of India. https://ncrb.gov.in/sites/default/files/CII-2021/CII_2021Volume%201.pdf.

National Family Health Survey. 2023. *National Family Health Survey, India*. https://www.nfhsiips.in/nfhsuser/aboutus.php.

National Literacy Mission Authority. 1988. Ministry of Education, Government of India. https://www.education.gov.in/nlma.

National Sample Survey Office (NSSO). 2023. *Annual Report 2022 - Periodic Labour Force Survey(PLFS)* (p. 758). (2023). Government of India, Ministry of Statistics and Programme Implementation, National Sample Survey Office. https://www.mospi.gov.in/sites/default/files/publication_reports/AR_PLFS_2022_23N.pdf.

Nieder, C., C. Muck, and J. Kärtner. 2019. "Sexual Violence against Women in India: Daily Life and Coping Strategies of Young Women in Delhi." *Violence Against Women* 25, no. 14: 1717–1738. https://doi.org/10.1177/1077801218824056.

Noughani, F., and J. Mohtashami. 2011. "Effect of Education on Prevention of Domestic Violence Against Women." *Iranian Journal of Psychiatry* 6, no. 2: 80–83.

O'Donnell, C. J., A. Smith, and J. R. Madison. 2002. "Using Demographic Risk Factors to Explain Variations in the Incidence of Violence Against Women." *Journal of Interpersonal Violence* 17, no. 12: 1239–1262. https://doi.org/10.1177/088626002237854.

Olson-Strom, S., and N. Rao. 2020. "Higher Education for Women in Asia." In *Diversity and Inclusion in Global Higher Education: Lessons From Across Asia*, edited by C. S. Sanger and N. W. Gleason, 263–282. Singapore: Springer Singapore. https://doi.org/10.1007/978-981-15-1628-3.

Panda, P. K. 2004. "Domestic Violence Against Women in Kerala." *Kerala Research Programme on Local Level Development, Centre for Development Studies* 86: 1–45.

Parayil, G. 1996. "The "Kerala Model" of Development: Development and Sustainability in the Third World." *Third World Quarterly* 17, no. 5: 941–957. https://doi.org/10.1080/01436599615191.

Priya, P., A. Venkatesh, and A. Shukla. 2021. "Two Decades of Theorising and Measuring Women's Empowerment: Literature Review and Future Research Agenda." *Women's Studies International Forum* 87: 102495. https://doi.org/10.1016/j.wsif.2021.102495.

Reddy, S., and S. S. Acharya. 2017. "Gender Violence in Delhi: Perceptions and Experience." In *Marginalization in Globalizing Delhi: Issues of Land, Livelihoods and Health*, edited by S. S. Acharya, S. Sen, M. Punia, and S.

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Reddy, 181–206. Pune: Springer India. https://doi.org/10.1007/978-81-322-3583-5_10.

Ross, K. 2006. "Status of Women in Highly Literate Societies: The Case of Kerala and Finland." *Literacy* 40, no. 3: 171–178. https://doi.org/10.1111/j.1467-9345.2006.00442.x.

Rowlands, J. 1995. "Empowerment Examined." *Development in Practice* 5, no. 2: 101–107. https://doi.org/10.1080/0961452951000157074.

Rowlands, J. 1998. "A Word of the Times, but What Does It Mean? Empowerment in the Discourse and Practice of Development." In *Women and Empowerment: Illustrations From the Third World*, edited by H. Afshar, 11–34. London Borough of Camden: Palgrave Macmillan UK. https://doi.org/10.1007/978-1-349-26265-6_2.

Roy, A. 2019. "Domestic Violence and Its Consequences on Women's Health: A Study Reference to Namasudra Community in Dakshin Dinajpur District, West Bengal." *Ensemble: A Bi-Lingual Peer Reviewed Academic Journal* 1, no. 2: 19–30.

Roy, A., and P. Chouhan. 2021. "Girl Child Marriage in Malda District of West Bengal: Analysis of Prevalence and Socio-Economic Factors." *Vulnerable Children and Youth Studies* 16, no. 4: 293–306. https://doi.org/10.1080/17450128.2021.1923878.

Roy, S., A. Bose, and I. R. Chowdhury. 2023. "Darkness Under the City Lights: A Qualitative Study on the Life of Urban Homelessness, Evidence From Siliguri City of West Bengal, India." *GeoJournal* 88, no. 2: 2263–2285. https://doi.org/10.1007/s10708-022-10753-x.

Saravag, P. K., and B. R. Kumar. 2024a. "A Hybrid Machine Learning and Regression Approach for Validating a Multi-Dimensional Crime Index in the Context of Crime against Women." *IEEE Access* 12: 117955–117969. https://doi.org/10.1109/ACCESS.2024.3439721.

Saravag, P. K., and R. B. Kumar. 2024b. "An Application of Scan Statistics in Identification and Analysis of Hotspot of Crime Against Women in Rajasthan, India." *Applied Spatial Analysis and Policy* 17, no. 3: 963–982. https://doi.org/10.1007/s12061-024-09572-z.

Sarva Shiksha Abhiyan Scheme. 2001. *Department of School Education & Literacy, Ministry of Education, Government of India*. https://www.india.gov.in/official-website-sarva-shiksha-abhiyan-0.

Sharma, A., A. Kela, S. Anchan, and P. Rojh. 2020. Male Literacy Rates & Crime Against Women: Regression Analysis (SSRN Scholarly Paper 3693734). Rochester, NY: SSRN. https://doi.org/10.2139/ssrn.3693734.

Sharma, K., M. Vatsa, M. Kalaivani, and D. Bhardwaj. 2019. "Mental Health Effects of Domestic Violence Against Women in Delhi: A Community-Based Study." *Journal of Family Medicine and Primary Care* 8, no. 7: 2522–2527. https://doi.org/10.4103/jfmpc.jfmpc_427_19.

Sharma, M., and S. Kumar. 2020. "Geographical Appraisal of Gender Disparity and Progress in Literacy of Haryana, India." *Indonesian Journal of Geography* 52, no. 2: 280–289.

Simister, J. 2011. "Assessing the 'Kerala Model': Education Is Necessary but Not Sufficient." *Journal of South Asian Development* 6, no. 1: 1–24. https://doi.org/10.1177/097317411100600101.

Simister, J. G. 2018. "Gender-Based Violence is Growing Problem in India." *Medical Research Archives* 6, no. 1: 1–21.

Tenkorang, E. Y., A. Y. Owusu, E. H. Yeboah, and R. Bannerman. 2013. "Factors Influencing Domestic and Marital Violence Against Women in Ghana." *Journal of Family Violence* 28, no. 8: 771–781. https://doi.org/10.1007/s10896-013-9543-8.

The Right of Children to Free and Compulsory Education Act. 2009. Government of India, Ministry of Education, Department of School Education and Literacy. https://dsel.education.gov.in/rte.

Verma, A., H. Qureshi, and J. Y. Kim. 2017. "Exploring the Trend of Violence Against Women in India." *International Journal of Comparative and Applied Criminal Justice* 41, no. 1–2: 3–18. https://doi.org/10.1080/01924036.2016.1211021.

Vicente, G., T. Goicoa, P. Fernandez-Rasines, and M. D. Ugarte. 2020. "Crime against Women in India: Unveiling Spatial Patterns and Temporal Trends of Dowry Deaths in the Districts of Uttar Pradesh." *Journal of the Royal Statistical Society Series A: Statistics in Society* 183, no. 2: 655–679. https://doi.org/10.1111/rssa.12545.

World Health Organization (WHO). 2023. *The Global Health Observatory, SDG Target 5.2 Eliminate Violence Against Women and Girls*. Geneva: World Health Organization. https://www.who.int/data/gho/data/themes/topics/sdg-target-5_2_1-eliminate-violence-against-women-and-girls.

Women and Men in India 2023—A statistical compilation of Gender related Indicators in India. 2023. Government of India, Ministry of Statistics and Programme Implementation, National Statistical Office, Social Statistics Division. https://www.mospi.gov.in/sites/default/files/reports_and_publication/statistical_publication/Women_Men/mw23/CompetePublication_WM2023.pdf.

Women and Men in India—A statistical compilation of Gender related Indicators in India—2021. 2022. Government of India, Ministry of Statistics and Programme Implementation, National Statistical Office, Social Statistics Division. https://www.mospi.gov.in/sites/default/files/reports_and_publication/statistical_publication/Women_Men/mw21/Women%20and%20Men%20in%20India%202021.pdf.

Xie, M., K. Heimer, and J. L. Lauritsen. 2012. "Violence against Women in US. Metropolitan Areas: Changes in Women's Status and Risk, 1980–2004*." *Criminology* 50, no. 1: 105–143. https://doi.org/10.1111/j.1745-9125.2011.00245.x.

Yodanis, C. L. 2004. "Gender Inequality, Violence Against Women, and Fear: A Cross-National Test of the Feminist Theory of Violence against Women." *Journal of Interpersonal Violence* 19, no. 6: 655–675. https://doi.org/10.1177/0886260504263868.

14682451, 2025, 256, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/issj.12550, Wiley Online Library on [19/01/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License